Harsh Desai

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Experimental systems engineer in low-pressure plasma, vacuum/HV test equipment, automation, detector calibration, and ADC/FPGA readout; targeting particle detector and plasma diagnostic roles.

EDUCATION

University of Michigan — **Ann Arbor, MI, USA**
Master of Engineering, Space Systems Engineering | GPA: 3.79/4.0 — Aug 2024–Dec 2025

Vellore Institute of Technology (VIT) — **Vellore, India**
Bachelor of Technology, Mechanical Engineering | GPA: 7.78/10 — Jul 2019–Jul 2023

TECHNICAL SKILLS

Plasma, Vacuum & Process Equipment: Low-pressure hollow-cathode plasma operation, 1.3 kV HV interface validation, vacuum chamber operation (CTI-Cryogenics Hi-Vac; roughing pump operated at 1 mTorr), RPA, emissive-probe and Langmuir-probe diagnostics, CEM/ESA calibration

Automation, Controls & DAQ: Python, LabVIEW (CLAD prep), MATLAB (learning), GPIB/IEEE-488, USB-serial, serial data logging, NI VISA (PyVISA, PyMeasure), Keithley DAQs and SMUs, TDK Lambda PSUs, High Voltage Switching Operations

FPGA, Digital Systems & Modeling: Verilog (learning), VHDL (learning), SIMION, SRIM, Amptek DPPMCA, FPGA Design Workflow, AMD Vivado, MATLAB HDL Coder & DSP Toolbox, SPENVIS, SWMF & Magnetosphere-Ionosphere Models (MAGE, SWMF, HYPERS), Google Data Studio

Mechanical & Manufacturing Tools: ANSYS (GUI/TUI): Fluent, Mechanical; PyANSYS, OptiSlang, OpenFOAM, SolidWorks, Fusion 360, AutoCAD, Autodesk Inventor, SpaceClaim, MIDO

RESEARCH & ENGINEERING EXPERIENCE

Climate and Space Sciences and Engineering, University of Michigan — **Ann Arbor, MI, USA**
Research Assistant, LVACCS Testing Rig Characterization — Feb 2026–Present

- Validated a 1300 V hollow-cathode plasma-source test workflow by combining HV discharge-box FMEA, Python/PyVISA ignition control, PSU logging, and DAQ synchronization; cut manual steps 15→5 and achieved 98.6% synchronized data
- Supported electrostatic regolith-lofting experiments using Retarding Potential Analyzer (RPA), emissive-probe, and Langmuir-probe diagnostics to compare plasma conditions across test states
- Characterized remote ignition and post-run processing workflows for Spacecraft Charging Device test readiness, improving experimental repeatability and saving ∼15 minutes of processing per run
- Built practical experience with low-pressure discharge operation, ignition stability, chamber diagnostics, and parameterized testing, creating a technical bridge toward plasma etch and thin-film deposition equipment

Space Physics Research Lab, University of Michigan — **Ann Arbor, MI, USA**
Summer Intern, TestBedz Spacecraft Qualification Platform — May 2025–Jul 2025

- Built a full-stack requirement-flowdown platform for TVAC, vibration, and EMI test submissions, matching spacecraft hardware profiles to facility limits across 6 test-type workflows
- Encoded hardware metadata, facility operating envelopes, and acceptance criteria into machine-readable compatibility checks and traceable test plans for repeatable equipment-qualification workflows

Solar and Heliospheric Research Group, University of Michigan — **Ann Arbor, MI, USA**
Student Research Assistant, SSD Readout Chain — Jan 2025–Dec 2025

- Designed an SSD readout roadmap around science-driven energy-resolution targets, integrating a Zmod ADC 1410 path with a Zynq-7000 architecture to evaluate a 1 MSPS→125 MSPS digitization upgrade
- Verified MATLAB HDL Coder golden models against Vivado co-simulations and diagnosed 125 MHz post-synthesis timing limits, including Worst Negative Slack constraints in the Zynq SoC signal path

L3Harris and University of Michigan — **Ann Arbor, MI, USA**
Communications Lead, Uranian Orbiter and Probe Mission Study — Sep 2024–May 2025

- Led Uranian mission communications architecture trades and DSN/fault-tolerance requirement flowdown, managing 15 traceable requirements and supporting a 16% ($450M) cost reduction through an orbiter-only configuration trade

SELECTED INSTRUMENTATION & PROCESS-RELEVANT PROJECTS

Space Instrumentation Calibration & Ion-Optics Series — Jul 2025–Dec 2025
Graduate Course: SPACE 571 | Supervisor: Prof. Stefano Livi

- Calibrated Channel Electron Multiplier response by mapping operating bias, pulse-height distributions, and amplifier gain behavior, connecting detector settings to usable signal quality and count-rate interpretation
- Calibrated the signal chain from charge injection through Charge-Shaper Amplifier, pulse shaping, and MCA acquisition, extracting gain, centroid, FWHM, and bias-dependent count response
- Modeled Electrostatic Analyzer and ion-optics performance using lab energy-angle scans, SIMION, and SRIM to quantify transmission, angular acceptance, energy resolution, FWHM, and filter response

PROJECTS & LEADERSHIP

CANSAT 2022 (NASA & AAS) | **Ranked 7th worldwide out of 42 teams** — Aug 2021–Jul 2022

- Built a 10 m tether-deployment, payload release, and camera-stabilization system using a DC motor, worm gear, torsion-spring latch, and dual-servo gimbal for controlled descent and fixed 45° imaging

Spaceport America Cup / IREC | **Placed 23rd globally and 5th in Asia-Pacific** — Apr 2020–Jul 2021

- Launched a Mach 0.9 solid-motor rocket to 10,000 ft and developed an SRAD trajectory simulator using CFD-backed aerodynamic data for flight prediction

CANSAT 2021 (NASA & AAS) | **Placed 13th globally and 7th in Asia-Pacific** — Aug 2020–Jul 2021

- Modeled maple-seed mono-wing payload descent using Blade Element Theory and transient CFD, with MATLAB telemetry software for real-time dual-payload deployment monitoring